



Evaluation of a Strategy Aiming to Enclose the Pulmonary Veins With Contiguous and Optimized Radiofrequency Lesions in Paroxysmal Atrial Fibrillation

A Pilot Study

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ABSTRACT

OBJECTIVES This study sought to evaluate the safety and the acute and 1 year outcomes of an ablation protocol aiming to enclose the PV with a contiguous and optimized RF circle by targeting region-specific criteria for lesion depth assessed by ablation index and interlesion distance.

BACKGROUND Reconnections after pulmonary vein (PV) isolation are explained by insufficient lesion depth and/or discontinuity of radiofrequency (RF) lesions.

METHODS A total of 130 consecutive patients with paroxysmal atrial fibrillation (AF) underwent PV encircling using a contact force-sensing catheter. RF was delivered targeting interlesion distance ≤ 6 mm and ablation index ≥ 400 at posterior wall and ≥ 550 at anterior wall. Recurrence was defined as any AF, atrial tachycardia (AT), or atrial flutter (AFL) (AF/AT/AFL > 30 s) on Holter electrocardiographs at 3, 6, and 12 months.

RESULTS Procedure and RF time per circle were 155 ± 28 min and 17 ± 5 min, respectively. Incidence of first-pass and adenosine-proof isolation were 98% and 98%, respectively. One short-lived transient ischemic attack was observed. At 12 months, single-procedure freedom from AF/AT/AFL was 91.3% in those 104 patients off antiarrhythmic drug therapy and 96.2% in those 26 patients on antiarrhythmic drug therapy. Single-procedure freedom from both AF/AT/AFL and antiarrhythmic drug therapy was 73.1%.

CONCLUSIONS This study suggests that an ablation protocol respecting strict criteria for lesion depth and contiguity results in acute durable PV isolation followed by a high single-procedure arrhythmia-free survival at 1 year. A prospective, multicenter trial is ongoing. (J Am Coll Cardiol EP 2018;4:99–108) © 2018 by the American College of Cardiology Foundation.

Single-procedure freedom from atrial fibrillation (AF) after radiofrequency (RF) pulmonary vein isolation (PVI) is $\approx 70\%$ in patients with paroxysmal AF (1–3). PV reconnection (PVR) is the major determinant of AF recurrence during follow-up (4–6). The use of contact force (CF)-sensing catheters, allowing better control over energy delivery during point-by-point pulmonary vein (PV)

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All authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the JACC: Clinical Electrophysiology [author instructions page](#).

Manuscript received April 28, 2017; revised manuscript received June 16, 2017, accepted June 26, 2017.

ABBREVIATIONS AND ACRONYMS

ADT	= antiarrhythmic drug therapy
AF	= atrial fibrillation
AFL	= atrial flutter
AI	= ablation index
ALCI	= ablation line contiguity index
AT	= atrial tachycardia
AU	= arbitrary unit
CF	= contact force
FTI	= force-time integral
ILD	= interlesion distance
IQR	= interquartile range
PV	= pulmonary vein
PVI	= pulmonary vein isolation
PVR	= pulmonary vein reconnection
RF	= radiofrequency

encircling, seems to improve freedom from AF to $\approx 80\%$ (7-10). Nevertheless, acute and late PVR still occur, and data on 1-year outcome are not consistent across studies (11,12).

We recently showed that acute and late PVR after CF-guided PVI are explained by lack of contiguity and insufficient lesion depth, as assessed by ablation index, within the deployed RF circle (13-15). Ablation index is based on the experimental work of Nakagawa et al. (14). This index, incorporating CF, power, and time in a weighted formula, predicted lesion depth in the canine ventricle with high accuracy.

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The CLOSE protocol is a new approach aiming to enclose the PV with contiguous and optimized RF lesions by targeting an interlesion distance (ILD) ≤ 6 mm and ablation index (AI) ≥ 400 at the posterior wall and ≥ 550

at the anterior wall. We evaluated 1) the safety; and 2) the acute and 1-year single-procedure outcomes after CLOSE-guided PVI in 130 patients with paroxysmal AF.

METHODS

STUDY SUBJECTS. From August 2013, all patients undergoing CF-guided ablation for AF at the St. Jan Hospital Bruges are followed in a prospective database approved by the local institutional review committee. This database implies collecting patients' written informed consent, a detailed case report form of the procedure, and follow-up with Holter electrocardiographs (ECG) at 3, 6, and 12 months. In this paper, we present the analysis of 130 consecutive cases of PVI for paroxysmal AF using the CLOSE protocol.

CLOSE-GUIDED PVI. PVI was performed by 4 operators under conscious sedation or general anesthesia. In anesthetized patients, esophageal temperature monitoring (SensiTherm, St. Jude Medical, St. Paul, Minnesota) was performed. After transseptal puncture (SLO, St. Jude Medical), a Lasso catheter and CF catheter (Thermocool SmartTouch, Biosense-Webster, Diamond Bar, California) were positioned in the left atrium and calibration of CF and respiratory gating were performed. Then we predefined the RF circle around the PV (nephroid shape) on the 3-dimensional geometry (Carto System, Biosense Webster). Point-by-point RF delivery was performed during sinus rhythm aiming for a contiguous circle enclosing the veins. Real-time automated display of RF applications (Visitag, Biosense Webster) was used with predefined

settings of catheter stability (3 mm for 8 s) and minimum CF (30% of time > 4 g). RF was delivered (EP Shuttle ST-3077, Stockert, Freiburg, Germany) in power-controlled mode (without ramping) with 25 to 35 W (irrigation flow up to 30 cc/min). RF was delivered until an AI of ≥ 400 at the posterior wall/roof and ≥ 550 at the anterior wall (Figure 1). In case of dislocation, a new RF application reaching the AI target was applied. Maximal ILD between 2 neighboring lesions was ≤ 6 mm. In case of chest pain or intraesophageal T° rise $> 38.5^\circ\text{C}$ during posterior wall ablation, RF delivery was stopped at an AI of 300. In the absence of first-pass isolation (i.e., no isolation after completing the circle), touch-up ablation was delivered until PVI. After PVI, we waited for 20 min, after which time, adenosine (dose resulting in atrioventricular block) was given 4 times (with the Lasso in its corresponding position). In case of reconnection during waiting time or during adenosine, the site of reconnection was located and treated with touch-up ablation until PVI was resistant to subsequent adenosine challenge. In case of (pre-) procedural documentation of typical flutter, cavotricuspid isthmus ablation was performed.

OFFLINE ANALYSIS OF CRITERIA TO ASSESS INTEGRITY OF THE DEPLOYED RF CIRCLE. Each procedure was analyzed offline. For each RF tag within the circle, we determined time of application (s), median delivered power (W), impedance drop ($\Delta\text{-Imp}$, Ω), average CF (g), force-time integral (FTI) (g/s), and AI (arbitrary unit [AU]). Custom-made software was used to automatically determine the path of encircling, perimeter of the RF circle, ILD (center-to-center distance, mm), and ablation line contiguity index (ALCI). ALCI was developed as a criterion to algorithmically score the combination of contiguity and lesion depth (13).

To describe the overall quality for a given circle, we determined for each circle the median value of all analyzed ablation parameters. Four circles from 2 patients were not available for offline analysis.

To describe the weakest link in a given circle, we determined for each circle the minimal value for AI and the maximal value for ILD.

Follow-up. Complications were reported on the case report form and collected during follow-up. After ablation, anticoagulation and antiarrhythmic drug therapy (ADT) were continued. At 3 months, anticoagulation was continued according to stroke risk, whereas ADT was continued at the discretion of the treating physician. Clinical evaluation and ECG were performed at 1, 3, 6, and 12 months. Holter ECG was performed at 3 and 6 months (24 h) and 12 months (7 days) or in case of symptoms. Freedom from recurrence was defined as 1-year absence from

AF, atrial tachycardia, and atrial flutter (AF/AT/AFL) >30 s beyond a 3-month blanking period.

STATISTICAL ANALYSIS. Normality of data distribution was tested with Shapiro-Wilk test. Continuous variables are expressed as mean $\pm$ SD if normally distributed, medians with interquartile ranges (IQRs) if non-normally distributed, and dichotomous variables as percentages. Kaplan-Meier estimate with 95% confidence intervals ($\pm$ 1.96 SE) was used to assess freedom from AF during 12-month follow-up period. All statistical analyses were performed in SPSS Statistics version 24 (IBM Corporation, Armonk, New York).

RESULTS

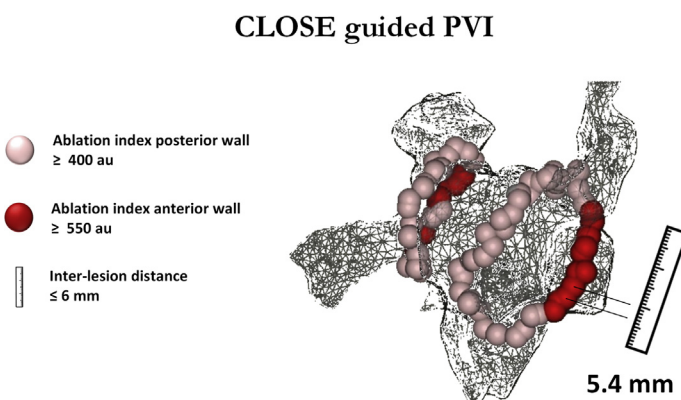
PATIENT CHARACTERISTICS. All 130 patients had paroxysmal AF. Baseline clinical characteristics are given in Table 1.

PROCEDURAL CHARACTERISTICS. Characteristics are given in Table 2. PVI was obtained in all cases. Cavotricuspid isthmus ablation was performed in 5 patients (3.8%). General anesthesia was used in 72 of 130 patients (55%). Procedure time was 155 ± 28 min (Figure 2A). RF time per circle was 17 ± 5 min (Figure 2B). Perimeter of the deployed circle was 115.7 ± 22.3 mm (Figure 2C). The number of RF tags to enclose the PV was 28 ± 5 (Figure 2D). The number of dislocations per circle was 3 ± 3 . Procedural characteristics for the right and left circles separately are given in Table 2.

Compared with conscious sedation, procedures with general anesthesia were characterized by shorter procedure time (147 ± 22 min vs. 164 ± 31 min; $p = 0.001$), shorter fluoroscopy time (16 ± 7 min vs. 17 ± 8 min; $p = 0.725$), lower dose area product value ($9,312 \pm 7,538$ mGy/cm² vs. $15,850 \pm 9,644$ mGy/cm²; $p = 0.014$) and shorter RF time for the right circle (15 ± 3 min vs. 18 ± 4 min; $p < 0.001$) and left circle (16 ± 4 min vs. 21 ± 7 min; $p < 0.001$).

RF CHARACTERISTICS AND INTEGRITY OF THE RF CIRCLE. A representative case illustrating the integrity of the RF circle is given in Figure 3. The CLOSE procedure is characterized by applications with relatively short duration (ranging from 11 to 61 s), a high rate of applications with high power (≥ 30 W), and a high rate of RF tags with a marked impedance drop ($\geq 7 \Omega$). There is a wide range of obtained CF values (5 g to 37 g) and FTI values (ranging from 184 to 875 g/s). Due to the nature of the protocol, applications at the anterior wall are all with an AI of ≥ 550 . Finally, the CLOSE-guided circle is characterized by enclosing lesions all with ILD ≤ 6 mm and ALCI of >100%.

FIGURE 1 Illustration of CLOSE-Guided PVI



See text for explanation. AU = arbitrary unit; CLOSE = enclosing the PV with contiguous and optimized RF lesions; ILD = interlesion distance; PVI = pulmonary vein isolation.

Overall, RF characteristics of lesion quality were as follows: median application time = 25 s (IQR: 23 to 35 s); median power = 35 W (IQR: 35 to 35 W); median obtained Δ -Imp = 12.7 Ω (IQR: 10.6 to 14.5 Ω); median CF = 15.0 g (IQR: 13.4 to 17.4 g); median FTI = 375 g/s (IQR: 326 to 446 g/s), median AI = 456 AU (IQR: 449 to 469 AU); median ILD = 4.1 mm (IQR: 3.8 to 4.4 mm); and median ALCI = 163 (IQR: 150 to 179).

Overall, weakest link analysis (Figures 2E and 2F) showed a median minimal AI of 408 AU (IQR: 378 to 425 AU) and 556 AU (IQR: 517 to 565 AU) at the posterior and anterior wall separately, whereas median maximal ILD was 6.5 mm (IQR: 5.9 to 8.0 mm).

Of interest, in 105 of 260 circles (40%), we did not target/reach the 400 AI value at the posterior wall because of chest pain and/or intraesophageal T° rise.

TABLE 1 Patient Characteristics (N = 130)

Paroxysmal AF	130 (100)
Age, yrs	59 $\pm$ 13
Male	84 (65)
BMI, kg/m ²	26.5 $\pm$ 4.1
Arterial hypertension	42 (32)
Structural heart disease	11 (9)
Diabetes	5 (4)
CHA ₂ DS ₂ VASc score ≥ 2	47 (36)
Left atrial diameter, mm	39.7 $\pm$ 3.7
Electrical cardioversion	26 (20)
First AF episode to PVI, months	24 (9-84)
Resistance to ADT	103 (79)

Values are n (%), mean $\pm$ SD, or median (interquartile range).

ADT = antiarrhythmic drug therapy; AF = atrial fibrillation; BMI = body mass index; CHA₂DS₂VASc = Congestive heart failure, Hypertension, Age ≥ 75 years, Diabetes mellitus, Stroke/transient ischemic attack/thromboembolism, Vascular disease, Age 65 to 74 years, Sex Category; PVI = pulmonary vein isolation.

TABLE 2 Procedure Characteristics (N = 130)

PV isolation	130 (100)
CTI ablation	5 (4)
General anesthesia	72 (55)
Procedure time, min	155 ± 28
Fluoroscopy time, min	16.7 ± 7.1
Dose area product, mGy/cm ²	11,314 ± 8,687
Total hospitalization time, h	39 ± 12
Right circles	
Perimeter, mm	123 ± 19
RF tags	29 ± 5
RF time, min	16 ± 4
First-pass isolation	130 (100)
Waiting/adenosine-proof isolation	128 (99)
Left circles	
Perimeter, mm	108 ± 23
RF tags	27 ± 5
RF time, min	18 ± 6
First-pass isolation	126 (97)
Waiting/adenosine-proof isolation	128 (99)

Values are n (%) or mean ± SD.
CTI = cavotricuspid isthmus; PV = pulmonary vein; RF = radiofrequency.

FIRST-PASS AND WAITING TIME/ADENOSINE-PROOF ISOLATION. Results are summarized in **Figures 2G and 2H**. Only 5 patients did not receive adenosine because of a history of asthma/chronic obstructive pulmonary disease. The CLOSE protocol was associated with high incidence of first-pass isolation (98%) and waiting time/adenosine-proof isolation (98%).

1-YEAR FREEDOM FROM AF/AT/AFL AFTER CLOSE-GUIDED PVI. Kaplan-Meier analysis was plotted in **Figures 4 and 5**. Overall, single-procedure 12-month freedom from AF/AT/AFL was 92.3% (**Figure 4**). Throughout the course of the study, 26 patients were on ADT. Those patients had continued taking ADT at the end of the blanking period. Conversely, no patient was taking ADT to treat a recurrence documented while being off ADT. In those 26 patients on ADT, freedom from AF/AT/AFL was 96.2% (**Figure 5**, lower left). In those 104 patients off ADT, freedom from AF/AT/AFL was 91.3% (**Figure 5**, upper left). Single-procedure freedom from both recurrence and ADT was 73.1%.

Single-procedure freedom from AF/AT/AFL was comparable among operators ($92.3 \pm 1.3\%$) and type of anesthesia (general = 90.3%, conscious sedation = 94.8%, $p = 0.511$).

Of interest, 16 patients (12%) showed early recurrence during the 3-month blanking period. Kaplan-Meier analysis for patients with ($n = 16$) and without ($n = 114$) arrhythmia during the blanking period is given in **Figure 5** (right).

SAFETY OF CLOSE-GUIDED PVI. One short-lived event, diagnosed as transient ischemic attack, occurred at day 8 after the ablation. Mean hospitalization length was 39 ± 12 h.

FINDINGS AT REPEAT ABLATION. All 10 patients with documented recurrence underwent repeat ablation. Of 10 patients, 9 revealed a normal bipolar voltage map during high-density mapping (>200 points). In 6 of 10 patients (60%), we observed a status of permanent isolation of all veins. In those 4 patients with reconnection of the veins, we observed 5 gaps in 4 of 8 circles. Relative to the RF characteristics of the first procedure, 1 gap could not be explained (adequate AI and ILD according to the CLOSE criteria), whereas 4 gaps were explained by a weak link in the initial ablation circle (all ILD >6 mm).

DISCUSSION

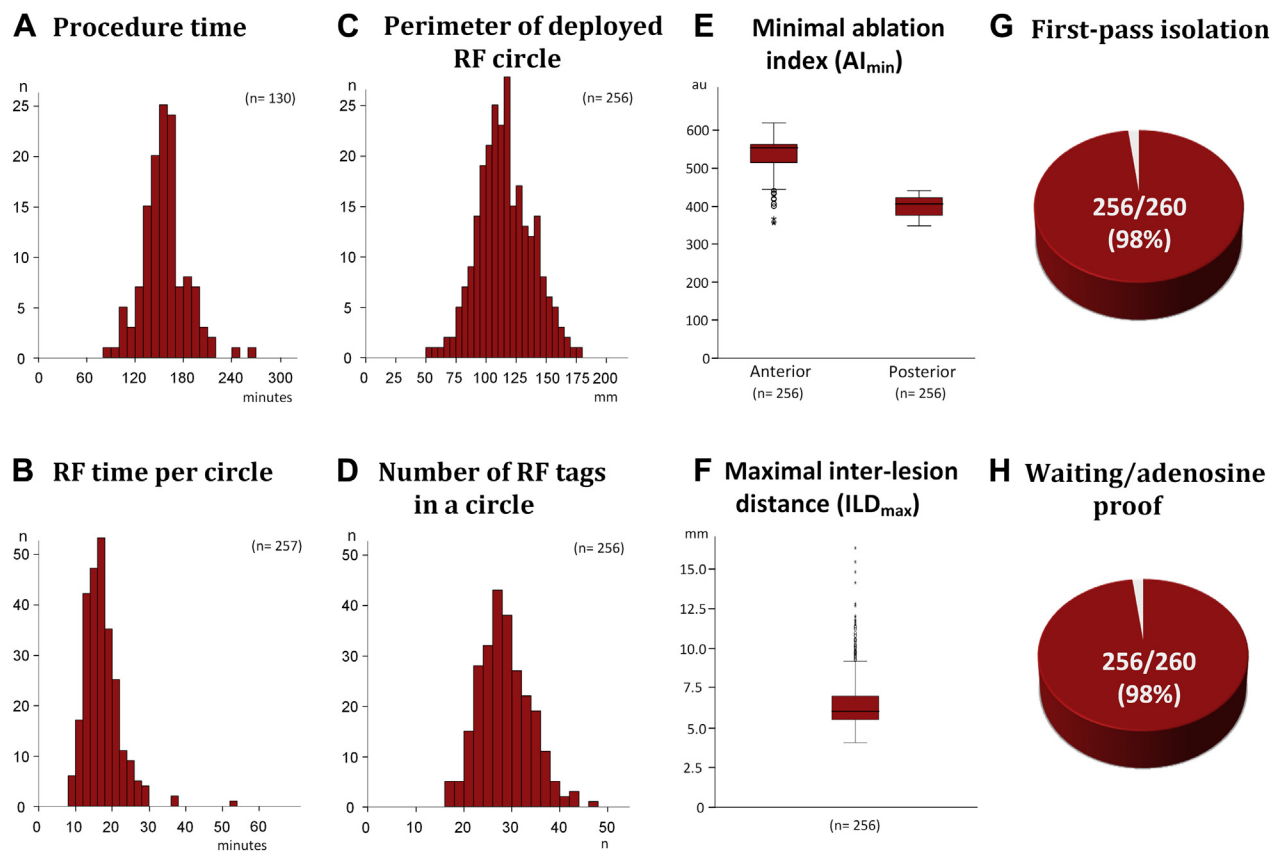
MAIN FINDINGS. This study, evaluating the CLOSE protocol in 130 patients shows that an ablation strategy aiming to enclose the PV with a contiguous ablation line with optimized RF lesions results in a high rate of adenosine-proof isolation. This is translated into a single-procedure arrhythmia-free survival of 91% at 1 year without ADT. These results are obtained without compromising safety and with relatively short procedure and RF times. These findings are in line with the hypothesis that avoiding weak links within the deployed RF circle is the key to durable PVI and clinical success (13).

SINGLE-PROCEDURE AF FREEDOM AFTER POINT-BY-POINT RF ABLATION. PVI is simultaneously the cornerstone and Achilles heel of ablation for paroxysmal AF. Despite acute isolation, late PVR results in AF recurrences with the ensuing need for repeat ablations. The recent FIRE AND ICE (FIRE AND ICE: Comparative Study of Two Ablation Procedures in Patients With Atrial Fibrillation) study confirmed a 64% single-procedure freedom from AF after point-by-point RF ablation (2,3).

Since the introduction of CF-sensing technology, several studies suggested enhanced acute durability of PVI and 1-year freedom from paroxysmal AF (7-10). In a parallel cohort study, CF guidance was associated with lower incidence of dormant conduction (8% vs. 35%) and better 1-year arrhythmia-free survival (88% vs. 66%) (7). In a single-center controlled study, CF guidance was associated with higher acute procedural success (80% vs. 36%) and 1-year AF freedom (90% vs. 67%) (8).

The above mentioned studies are in line with work from Reichlin et al. (9), showing 84% AF freedom if

FIGURE 2 RF Characteristics and Incidence of Acute Durable PVI



(A) Procedure time. (B) Radiofrequency (RF) time per circle. (C) Perimeter of the deployed circle. (D) Number of RF tags within the circle. (E) Obtained minimal ablation index (AI_{min}) per circle. (F) Obtained maximal ILD. (G) Incidence of first-pass isolation. (H) Incidence of waiting time/adenosine-proof isolation. See text for further explanation. Abbreviations as in Figure 1.

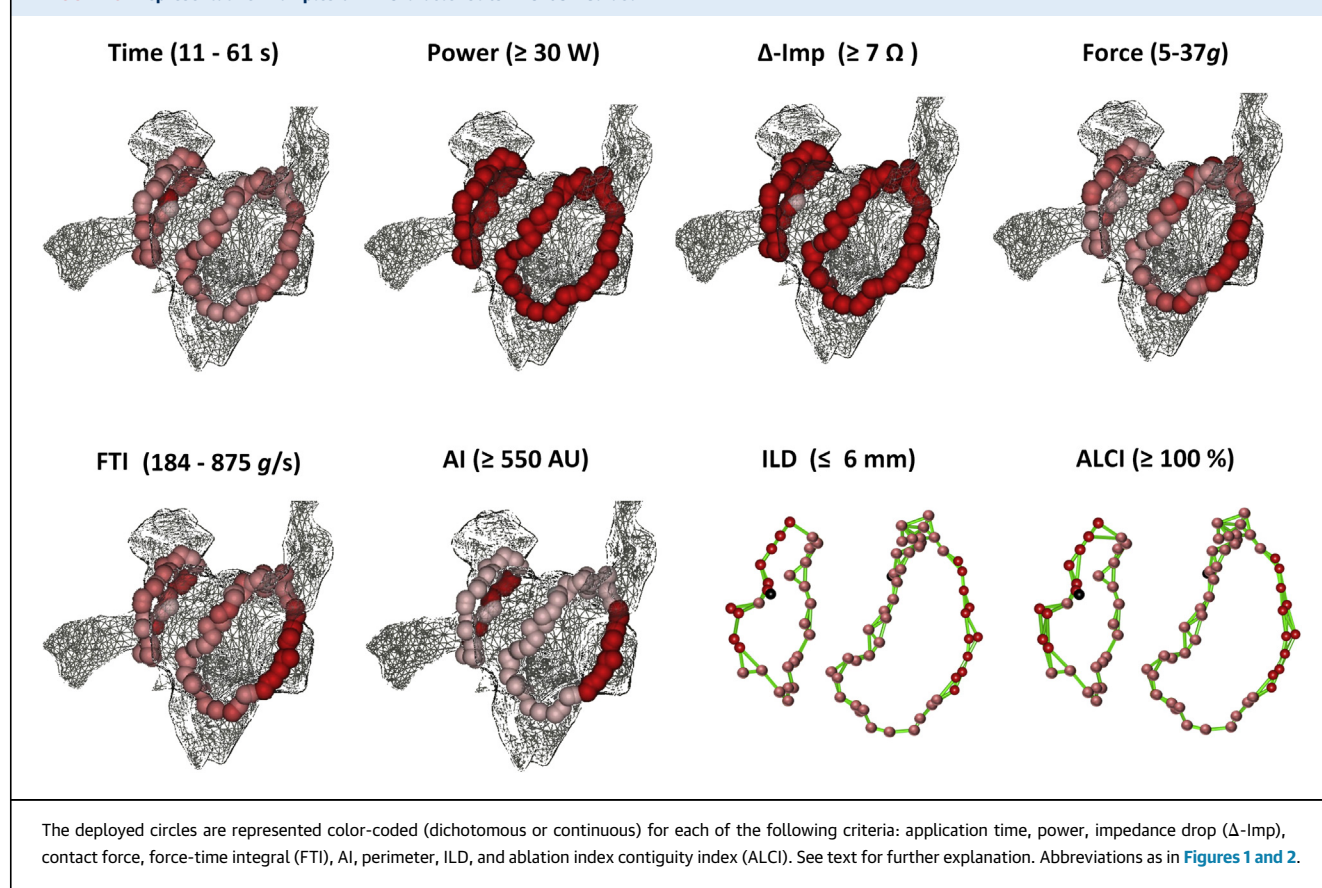
PVI is guided by initial impedance decrease, as an indicator of good catheter contact. Improved outcome is also in line with experimental data showing that for a given stable catheter position and for a given power, lesion depth and width, and volume increase with higher CF (16-20).

However, data on 1-year outcomes are not consistent across all studies (11,12). A recent multicenter randomized controlled study in 117 paroxysmal AF patients failed to demonstrate a beneficial effect of CF guidance on AF-free survival (11). Also Reddy et al. (12) failed to demonstrate a benefit of CF-guided ablation in a multicenter randomized trial. Only in the subgroup of patients with at least 90% of RF applications ≥ 10 g was AF freedom superior to nonoptimal CF (76% vs. 58% at 1 year).

On top of these conflicting results, even in the most promising scenario, still $\approx 20\%$ of the patients require a repeat procedure after CF-guided PVI.

DETERMINANTS OF GAP IN CF-GUIDED PVI. The conflicting results indicate that CF alone cannot preclude the presence of weak links in the ablation chain. Recently, our group demonstrated that in CF-guided PVI, acute and late PVR are explained by discontinuity (ILD) and/or insufficient lesion depth (AI) of the lesions within the deployed RF circle (13).

Prior studies already hinted to the importance of contiguous lesions to obtain acute durable isolation. Miller et al. (21) showed that closing the “visual” gap was essential for acute durable PVI, whereas Park et al. (22) showed that acute PVR despite CF ≥ 10 g was explained by ablation sites with an ILD of ≥ 5 mm. A recent subanalysis of SMART-AF (ThermoCool SmartTouch Catheter for Treatment of Symptomatic Paroxysmal Atrial Fibrillation) in 40 paroxysmal AF patients also showed that ILD correlates with success rate (1-year outcomes ranging from 33% to 93%) (23).

FIGURE 3 Representative Examples of RF Characteristics in CLOSE-Guided PVI

These findings support the concept of superiority of contiguous overlapping lesions with sufficient energy delivery. This concept is also in line with human cardiac magnetic resonance studies showing that acute isolation can be achieved by a combination of ablation-induced reversible (edema) and irreversible atrial injury (necrosis) (24). Likewise, experimental data showed that tissue heating can lead to acute changes in electrophysiological properties allowing for larger gaps to have conduction block, but as the tissue recovers, to become conductive again (25). This reversible atrial injury is more likely to occur at the periphery of the lesion (where the tissue temperature is lower), but it can also occur at the center of the ablation point when insufficient energy is delivered.

The key observation that lower ablation index values are associated with reconnection—even in segments characterized by closely spaced lesions—suggests that ablation index can be used to titrate delivery of sufficient energy (13). By taking into account the greater contribution of power (over CF) and of the initial time of ablation (26), AI differs from force-power-time integral calculated as the

multiplication of force, power, and time (27). Of interest, El Haddad et al. (13) observed that there is a significant difference in minimal AI values required to obtain durable segments in the anterior part of the circle compared with the posterior segments, suggestive of thicker atrial tissue in anterior regions.

Based on the findings by El Haddad et al. (13), we hypothesized that, to obtain durable PVI, maximum ILD should not exceed 6 mm and that minimal lesion depth defined by the AI should reach the value of at least 400 at the posterior wall and at least 550 at the anterior wall (the CLOSE criteria).

STRENGTHS AND WEAKNESSES OF THE CLOSE PROTOCOL. In the present relatively large patient cohort, we observed a high rate of acute durable isolation and a high 91% single-procedure, 1-year freedom from AF/AT/AFL without ADT. If continuing ADT (without evidence of recurrence) is considered as failure (1), single-procedure freedom from both arrhythmia recurrence and ADT was 73.1%. Also the 60% prevalence of permanent and complete PVI at repeat is at the higher end when compared with

prior studies reporting on the status of complete PVI in patients undergoing a repeat procedure for arrhythmia recurrence (4-6,28). Taken together, these data suggest that the CLOSE protocol results in acute and late durable PVI.

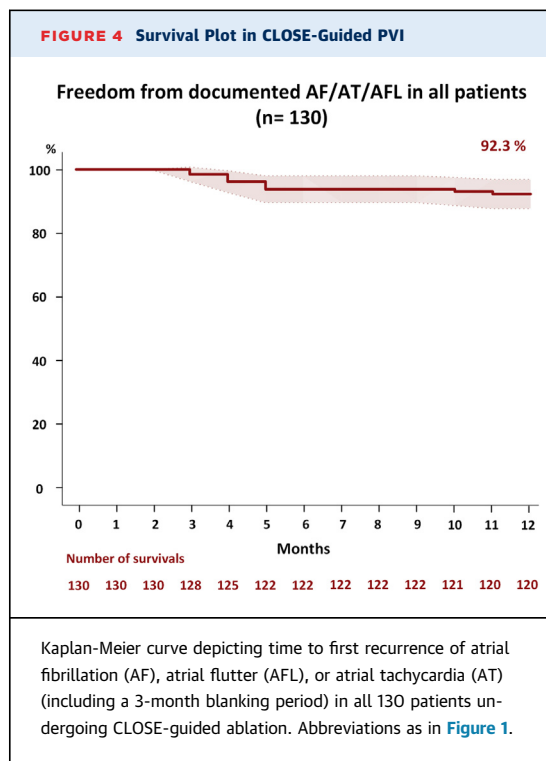
The CLOSE protocol describes a method to obtain PVI with objective and quantifiable criteria (AI, ILD, and ALCI). Although integrity of the deployed RF circle is expected to be similar when all parameters are met, reproducibility across centers requires further study.

We surmise that the CLOSE protocol is different from currently used “dragging” or point-by-point RF strategies. Although anecdotally some operators already pursue a contiguous ablation approach with overlapping lesions, this claim is not justified by objective measures. In the SMART-AF subanalysis—most likely reflecting expert use of CF-guided PVI—more than one-half of the patients were treated with RF circles characterized by any ILD of >10.6 mm (23). Lack of verification of ILD is reflected in the ongoing need for touch-up ablation after first encirclement or adenosine challenge.

Also the use of AI to optimize RF delivery is different from conventional CF- or FTI-guided strategies. In the CLOSE protocol, low CF values per se do not preclude delivery of RF at that site. In contrast to prior studies, obtained FTI values at the posterior wall are as low as 200 g/s (29,30). The use of higher power implies that lesions can be deployed with a relatively short application time (resulting in few dislocations), even at sites with low but stable contact. Finally, in contrast to FTI or CF targets, lesion quality is expected to be similar at a certain AI value (despite differences in power, force, and time of application required to reach that value) (27).

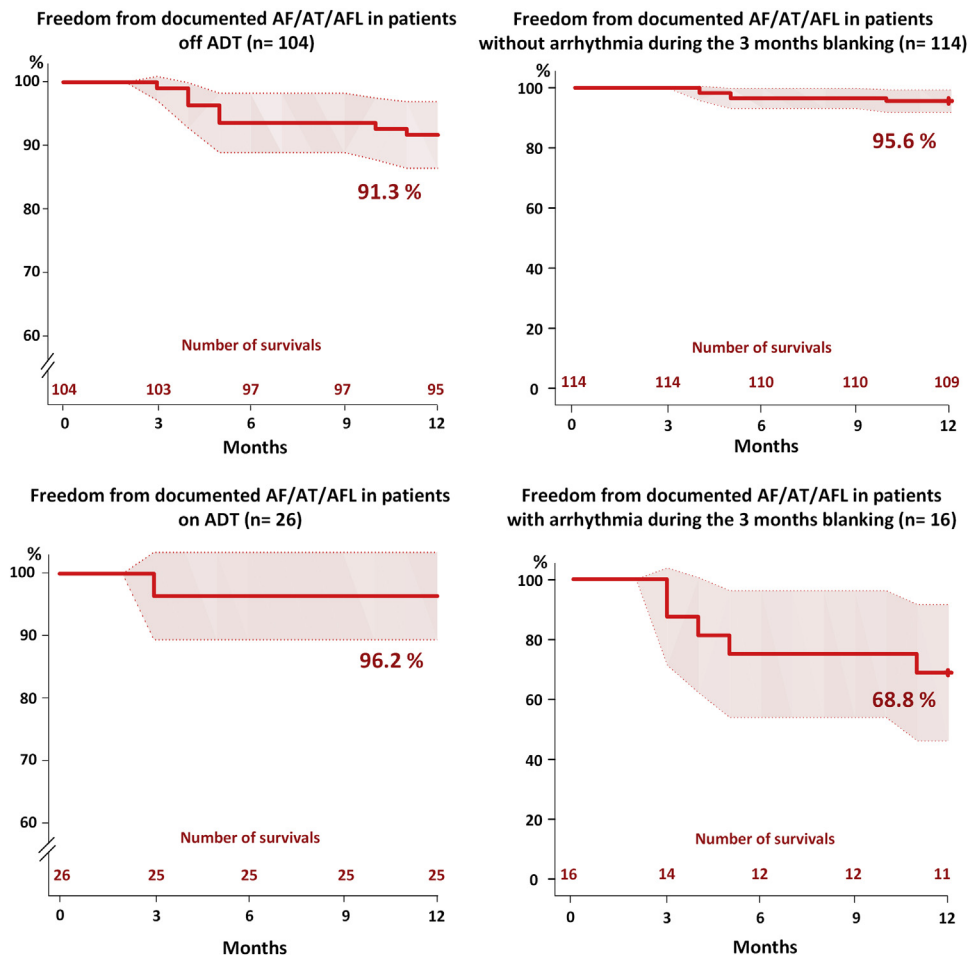
Our data suggest that durable PVI can be obtained with relatively short procedure and RF times. This is explained by the fact that AI allows use of higher power during RF delivery (shorter applications) and by the fact that no time is lost for chasing and managing acute PVR. Although our data suggest that procedural efficiency is higher using general anesthesia, outcome was found to be similar.

The present population is too small to draw any definitive conclusions regarding safety. There is no preclinical data to support the safety of any strategy using higher power RF delivery. Nevertheless, in those 130 patients undergoing the CLOSE protocol, we observed no complications, especially no audible steam pop, no signs of clinical PV stenosis, no cardiac perforation, no permanent



stroke, no atrioesophageal fistula, and no deaths. Absence of esophageal injury despite relatively high power can be explained by short application times at the posterior wall (maximum 20 s) and esophageal T° monitoring.

Finally, we retain considerable challenges for future RF-guided PVI (1). Our strategy requires validation and optimization using other ablation catheters and/or ablation parameters settings (lower irrigation, higher power, and so on) (2). Although we observed a limited number of dislocations during encircling, it remains to be seen whether this is reproducible across centers (3). The CLOSE protocol seems appropriate to obtain durable PVI using this strategy of deploying ≈30 RF lesions following a 10- to 12-cm path around the PV. Whether the CLOSE criteria (AI, ILD, and ALCI) might offer an ideal platform to guide wider encirclement of the PV requires further study (4). Cutoff-guided ablation strategies, as advocated in the present study, might result in durable isolation at the cost of overshoot and potential safety issue. The current observation of acute durability despite the delivery of AI values lower than 400 at the posterior wall might be a first step toward avoiding this overshoot. Moreover, AI is an estimate and not a direct measure of lesion depth. To minimize RF delivery, each application should be individualized according to online

FIGURE 5 Survival Plots for Patient Subgroups

Kaplan-Meier curves depicting time to first recurrence of AF, AFL, or AT in patients undergoing CLOSE-guided ablation for subgroup of patients not taking antiarrhythmic drug therapy (ADT) (**upper left**) or taking ADT (**lower left**), and without arrhythmia during the 3 months blanking (**upper right**) or with arrhythmia during blanking (**lower right**). Abbreviations as in [Figures 1 and 4](#).

in vivo characterization of the tissue before, during, and after RF delivery. Only then will the unique advantage of point-by-point ablation be exploited to its full potential.

STUDY LIMITATIONS. Although 130 patients were enrolled, several limitations need to be underlined.

First, as with many pilot studies, this study was monocentric. We believe, however, that this feasibility study in a relatively large cohort of 130 patients undergoing CLOSE-PVI was a first and necessary step toward adequately powered multicenter studies.

Despite a 24-h Holter ECG at 3 and 6 months and a 7-day Holter ECG at 12 months and the symptomatic

nature of PAF, the discontinuous nature of monitoring can underestimate AF recurrence.

In the present study, we did not perform voltage mapping at baseline. No preablation imaging with computed tomography or cardiac magnetic resonance was performed. Therefore, we cannot exclude the possibility of an aberrant PV.

Finally, ADT was not systematically withdrawn after the blanking period. Due to the large sample size, however, we could report on outcome in 104 CLOSE patients not taking ADT.

These limitations are being tackled by ongoing clinical studies (a multicenter trial and a study using subcutaneous loop recorders) evaluating the safety and efficacy of the CLOSE protocol.

CONCLUSIONS

In the present study, we evaluated the feasibility of an ablation protocol aiming to enclose the PV with contiguous and optimized RF lesions. We observed a relatively short procedure time, high rate of acute PVI durability, and a high single-procedure arrhythmia-free survival at 1 year.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: PVR is caused by insufficient lesion depth and/or discontinuity within the deployed RF circle. This pilot study shows that an ablation strategy aiming to enclose the pulmonary veins targeting strict criteria of AI and ILD results in a high acute PVI durability and high single-procedure arrhythmia-free survival at 1 year.

TRANSLATIONAL OUTLOOK: Integration of electroanatomic mapping with region-specific criteria for AI and ILD may create durable lesions and decrease the need for repeat procedures.

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KEY WORDS atrial fibrillation, CLOSE protocol, contact force, contiguity, lesion depth, pulmonary vein isolation